GIGABYTE AMD Radeon RX 9070 GRE OC

The Sweet Spot that AMD Needed

6.7

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The AMD Radeon RX 9070 GRE enters the stack as a very deliberate middle child. It sits below the Radeon RX 9070 and RX 9070 XT, but above the more mainstream Radeon RX 9060 XT, and its geared to deliver strong 1440p gaming without pushing the buyer all the way into higher-end territory. In AMD's own positioning, the RX 9070 GRE is meant to sit closer to the GeForce RTX 5070 in raster performance while being priced closer to the GeForce RTX 5060 Ti 16GB, which makes value the real argument here. Positioning in India might go in an entirely different direction though.

For the review, we picked cards that are within the pricing vicinity of the RX 9070 GRE. Against the RX 9070 and RX 9070 XT, the GRE shows how much of the RDNA 4 experience AMD can retain at a lower tier. Against the older RX 7900 GRE, it tells us what RDNA 4 changes in practice, especially for ray tracing and AI-assisted features. Looking at the competition i.e. NVIDIA's RTX 5070 and RTX 5060 Ti, it becomes a more direct fight around 1440p gaming, memory configuration, driver features, power behaviour, and upscaling support. The RX 9070 GRE is trying to be the most sensible card in this stack for high-refresh 1440p gaming. At a price point of USD 599, the AMD Radeon RX 9070 GRE is priced to replace the earlier RX 9070 but we still haven't received Indian pricing, so you'll have to wait another day to see where it lies among the competition.

PERFORMANCE

The comparison set for this review includes the Radeon RX 9070 GRE, Radeon RX 9070, Radeon RX 9070 XT, Radeon RX 7900 GRE, GeForce RTX 5070, Radeon RX 9060 XT, and GeForce RTX 5060 Ti. This gives the RX 9070 GRE a useful set of references: its direct RDNA 4 siblings, the previous-gen GRE card, and NVIDIA's closest current rivals. The key question is whether the RX 9070 GRE behaves like a cut-down enthusiast card or a stretched mainstream GPU. In rasterised games, the card is expected to sit closer to the RTX 5070 than the RTX 5060 Ti.

- Processor – AMD Ryzen 7 9800X3D
- CPU-Cooler – Noctua NH-D15
- RAM – 2x 24 GB Kingston FURY Beast 6000 MT/s
- SSD – Kingston KC3000 2TB
- PSU – Cooler Master V1200

3DMark

3DMark is a popular benchmarking tool for graphics cards and gaming systems. It provides a comprehensive suite of tests that measure various aspects of a GPU's performance. 3DMark includes several benchmarks, such as Time Spy, Steel Nomad, Speed Way, and Fire Strike. Time Spy is a DirectX 12 benchmark that tests the performance of a GPU and CPU in a variety of graphically demanding scenes. Steel Nomad, successor to Time Spy, is a DirectX 12 benchmark that focuses on real-time ray tracing and other advanced graphics features. Then there's Speed Way, a DirectX 12 benchmark designed to test the performance of a GPU and CPU in a high-speed racing game environment. And for legacy benchmarks, we have Fire Strike which is a DirectX 11 benchmark that tests the performance of a GPU and CPU in a variety of gaming scenes.

Ray Tracing

3DMark Port Royale is a synthetic benchmark that uses a real-time ray tracing scene to simulate the reflections, shadows, and visual effects that are possible with ray tracing tech. Port Royale is a demanding benchmark that can be used to compare the performance of all current graphics cards with real-time hardware-accelerated ray-tracing.

OpenCL Rendering

This benchmark utilises OpenCL to generate photorealistic results by adhering to the physics of light, a process known as physically-based rendering. Rendering progress is gauged by the number of samples calculated, which can be visualised as "light particles" that interacted with the scene and reached the camera's sensor. As a physically-based renderer, the results reflect how GPUs are used in industrial rendering applications.